



Manish Singh

Master of Technology
in Computer Science Engineering
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EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Master of Technology	Indian Institute of Technology, Ropar	8.3	2025-2027
Bachelor of Technology	Haldia Institute of Technology, West Bengal	9.06	2017-2021
Senior Secondary	Central Board of Secondary Education	91.6%	2016
Secondary	Central Board of Secondary Education	10	2014

EXPERIENCE

- Intel** Jul 2025 - Present
AI Software Engineer Graduate Intern Bengaluru
 - Building multiple internal Model Context Protocol (MCP) servers — including one for natural language-to-SQL translation and automated data retrieval — designed to work together as tools for a single agent
 - Working on agentic AI framework development at Intel, integrating tool-calling and multi-step orchestration to enable AI-powered assistants across internal systems and workflows.
- Infosys** Aug 2021 - Oct 2024
Specialist Programmer Pune
 - Engineered Spark analytical pipelines in Azure HDInsight, orchestrating data processing through Azure Data Factory (ADF); extracted actionable insights from datasets, reducing processing time by 30% and boosting data reliability.
 - Optimized pipeline performance, resulting in a notable 25% reduction in total processing time.
 - Developed robust CI/CD pipelines using Azure DevOps; reduced deployment time by 40% and increased release frequency from monthly to bi-weekly, enhancing overall efficiency.
 - Technologies: Azure HDInsight, Azure DataFactory, Azure DevOps, Spark, Scala, Azure SQL Database

PROJECTS

- State-wise YouTube Trend and Sentiment Analysis using NLP and BDA** Jan 2026 - May 2026
Course GitHub
 - Developed a system to analyze YouTube trends and sentiment at a state level by extracting topics, mapping them to regions, and visualizing insights via a dashboard.
 - Built the NLP pipeline including multilingual preprocessing, semantic topic modeling (Sentence-BERT + KeyBERT), sentiment analysis (VADER), and integration with PySpark for scalable processing.
 - Tech Stack: Python, PySpark, Sentence-BERT, KeyBERT, VADER, FLAN-T5, Delta Lake, Streamlit, Docker
- Multi-Paradigm Content-Based Image Retrieval System** Month Year – Month Year
Course GitHub
 - Proposed and implemented MSTE-RG, a custom 76-dimensional image descriptor fusing multi-scale texture statistics, spatial edge density, and DCT coefficients with a self-designed Radial Gradient Energy Profile to separate central subjects from background clutter.
 - Benchmarked the novel descriptor against five retrieval paradigms (LBP, handcrafted color features, shallow/deep NN embeddings, pretrained ResNet-18) on CIFAR-10, MNIST, and Fashion-MNIST, then designed a hybrid score-level fusion combining MSTE-RG with CNN features, evaluated via Precision@K, Recall@K, and mAP.
 - Tech Stack: Python, PyTorch, Torchvision, scikit-learn, scikit-image, Gradio
- Campus Court - Sports Facility Management App** July 2025 - Nov 2025
Course GitHub
 - Engineered a full-stack Android application for real-time booking of campus sports facilities and student team management at IIT Ropar.
 - Implemented secure authentication and live slot availability using cloud-hosted APIs, supporting concurrent users and conflict-free bookings.
 - Tech Stack: Django (REST Framework), Redis (Caching), Supabase (PostgreSQL), Flutter, Render (Cloud Hosting)

TECHNICAL SKILLS

- Programming Languages:** C/C++, Python, SQL
- Machine Learning & AI:** Machine Learning, Deep Learning, Computer Vision, PyTorch, LangGraph, RAG, Vanna AI, MCP
- Backend & Databases:** Django, PostgreSQL
- Big Data & Cloud:** Spark, PySpark, Kafka, Microsoft Azure, Azure Data Factory, Databricks
- DevOps & Tools:** Docker

KEY COURSES TAKEN

- CSE & Maths:** Data Structure & Algorithm, Operating System, DBMS, Computer Systems, Mathematical Foundation for Computer Science, Internet of Things (IoT), Machine Learning, Deep Learning for Computer Vision

ACHIEVEMENTS

- Received outstanding performance rating for exemplary project contributions and dedication 2022
- INSTA Award, Certificate of Appreciation for Excellent Performance in a project from Infosys 2022